

Hye-Young Jo Curriculum Vitae

Roser ATLAS Center, Room 231
1125 18th St, 320 UCB
Boulder, CO 80309-0320, United States

www.hyeyoungjo.com
hye-young.jo@colorado.edu
[Google Scholar](#)

Research Interests

I am a Ph.D. student in Computer Science at the University of Colorado Boulder. My main research area is **Human-Computer Interaction**. My research goal is to **expand people's capacity to create and learn** by building interactive systems with generative AI and extended reality that adapt content to their actions, context, and intentions. My work spans AI-assisted filmmaking, technology-enhanced learning, and embodied fitness training.

Keywords: human-AI interaction, creativity support tools, adaptive media, embodied interaction

Education

Summary: I trained in fine art, then industrial design, and now study computer science.

August 2024 –
Present **University of Colorado Boulder**
Ph.D. student, Computer Science
Supervisor: Ryo Suzuki

September 2020 –
August 2022 **Korea Advanced Institute of Science and Technology (KAIST)**
Master of Science, Industrial Design, *Best Master's Thesis Award*
Thesis: Exploring Different Augmented Reality Visualization Methods of Fitness Videos and Their Effects on the Exercise Experience
Supervisor: Andrea Bianchi

March 2011 –
February 2016 **Seoul National University**
Bachelor of Fine Arts, Painting and Media Arts, *Cum Laude*
Painting graduation exhibition on fluid identity [A1]
Media Arts graduation exhibition on Korea's [Sampo generation](#) [A4]
Supervisor: Inhwon Oh, Cheol-Woong Sim

Research Experience

Summary: I have built interactive systems in five research groups across academia and industry, including research internships at Autodesk and Fujitsu.

May 2026 –
August 2026 [Autodesk, HCI and Visualization Team](#)
Research Intern, Mentors: Frederik Brudy, David Ledo
Created a design space for film directing, grounded in a cinematic corpus, to establish a shared language between human creators and generative AI.

May 2025 –
October 2025 [Fujitsu Research of America, Converging Lab](#)
Research Intern, Mentors: Mose Sakashita, Aditi Mishra, Aakar Gupta, Koichiro Niinuma
Built a system that grounds AI video generation in street-view imagery for spatially consistent shots, evaluated with 12 filmmakers [C8].

August 2024 – [University of Colorado Boulder, Programmable Reality Lab](#)

Present

Graduate Research Assistant, Mentor: Ryo Suzuki

Study how people create and consume video by building interactive systems with generative AI, extended reality, and virtual agents for storytelling and education [C7, C8, U1].

January 2023 – [Chung-Ang University, Human-AI Interaction Design & Fabrication Lab](#)

July 2024

Research Assistant, Mentor: Yoonji Kim

Built a filmmaking previsualization tool using image segmentation and deepfake face swapping [C4], and haptic systems that recreate a personal trainer's touch through vibrotactile and electrical stimulation [P1, C6].

September 2020 – [KAIST, Make Lab](#)

September 2022

Graduate Research Assistant, Mentor: Andrea Bianchi

Designed and evaluated AR, VR, and haptic systems for home fitness, remote education, and immersive VR [C1-C3].

Work Experience

Summary: Before research, I spent nearly four years at Dexter Studios, first as a film VFX compositor and then as a VR/AR generalist.

August 2018 – **VR/AR Generalist at [Dexter Studios](#)**

April 2020

New Media Division, Immersive Content and Virtual Production

Created 3D graphics for VR films, AR exhibitions and apps, and virtual production, covering character design, environments, visual scripting, lighting, and look development [X1-X2, X4].

September 2016 – **Film VFX Compositor at [Dexter Studios](#)**

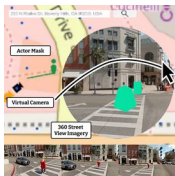
July 2018

Film Post-Production VFX Compositing Team

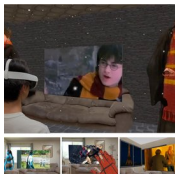
Composited live-action plates, 2D elements, and 3D CG renders into seamless final shots for feature films [F1-F6].

Peer-Reviewed Full Papers

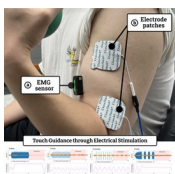
Summary: 8 full papers published in CHI (4), UIST (2), ISMAR (1, in IEEE TVCG), and HCIK (1). Two additional full papers are under review.



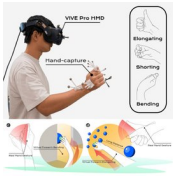
[C8] Hye-Young Jo, Mose Sakashita, Aditi Mishra, Ryo Suzuki, Koichiro Niinuma, Aakar Gupta. **Map2Video: Guiding Real-World-Grounded AI Video Generation**. In Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2026. (UIST '26, acceptance rate: 20.1%) Also accepted to the UIST '26 demo track. ([video](#)) ([PDF](#))



[C7] Keiichi Ihara, DaeHo Lee, Manato Abe, Hye-Young Jo, Ryo Suzuki. **CinemaWorld: Generative Augmented Reality with LLMs and 3D Scene Generation for Movie Augmentation**. In Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2026. (UIST '26, acceptance rate: 20.1%) Also accepted to the UIST '26 demo track. ([PDF](#))



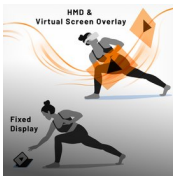
[C6] Dong-Uk Kim, Hye-Young Jo, Hankyung Kim, Ryo Suzuki, Seungwoo Je, Yoonji Kim. **TingleTouch: Touch Guidance through Electrical Stimulation in Resistance Training**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2026. (CHI '26, acceptance rate: 25.3%) ([video](#)) ([PDF](#))



[C5] Yilong Lin, Han Shi, Weitao Jiang, Xuesong Zhang, **Hye-Young Jo**, Yoonji Kim, Seungwoo Je. **Designing Hand and Forearm Gestures to Control Virtual Forearm for User-Initiated Forearm Deformation**. IEEE Transactions on Visualization and Computer Graphics. IEEE, 2025.
(ISMAR '25, acceptance rate: 7.86%) ([video](#)) ([PDF](#))



[C4] **Hye-Young Jo**, Ryo Suzuki, Yoonji Kim. **CollageVis: Rapid Previsualization Tool for Indie Filmmaking using Video Collages**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2024.
(CHI '24, acceptance rate: 26.4%) ([video](#)) ([PDF](#))



[C3] **Hye-Young Jo**, Laurenz Seidel, Michel Pahud, Mike Sinclair, Andrea Bianchi. **FlowAR: How Different Augmented Reality Visualizations of Online Fitness Videos Support Flow for At-Home Yoga Exercises**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2023.
(CHI '23, acceptance rate: 27.6%) ([video](#)) ([PDF](#))



[C2] **Hye-Young Jo**, Wooje Chang, Hoonjin Jung, Andrea Bianchi. **Design of Virtual Reality Application for Interaction Prototyping Remote Education**. In Proceedings of HCI Korea. The HCI Society of Korea, 2022.
(HCIK '22) **Best Paper Award** ([video](#)) ([PDF](#))



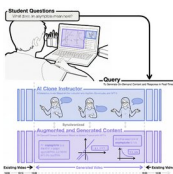
[C1] Neung Ryu, **Hye-Young Jo**, Michel Pahud, Mike Sinclair, Andrea Bianchi. **GamesBond: Bimanual Haptic Illusion of Physically Connected Objects for Immersive VR Using Grip Deformation**. In Proceedings of the CHI Conference on Human Factors in Computing Systems. ACM, 2021.
(CHI '21, acceptance rate: 26.3%) **Honorable Mention Award (top 5%)** ([video](#)) ([PDF](#))

Posters



[P1] **Hye-Young Jo**, Chan Hu Wie, Yejin Jang, Dong-Uk Kim, Yurim Son, Yoonji Kim. **TrainerTap: Weightlifting Support System Prototype Simulating Personal Trainer's Tactile and Auditory Guidance**. In Adjunct Proceedings of the Annual ACM Symposium on User Interface Software and Technology. ACM, 2023.
(UIST '23 Adjunct, acceptance rate: 21.0%) ([video](#)) ([PDF](#))

Preprints



[U1] **Hye-Young Jo**, Ada Zhao, Xiaolan Liu, Ryo Suzuki. **Generative Lecture: Making Lecture Videos Interactive with LLMs and AI Clone Instructors**. arXiv preprint. 2025.
([video](#)) ([PDF](#))

Fellowship and Awards

Summary: \$176,000 in total funding awarded.

- Fall 2025 –* **Google Ph.D. Fellowship**
Spring 2027 Awarded in the Human-Computer Interaction category; full tuition and stipend (\$85,000/year).
- 2026 **Conference Support Fund**
Awarded by the Department of Computer Science, University of Colorado Boulder (\$1,500).
- 2025 **Ralph J. Slutz Student Excellence Award**
Awarded by the Department of Computer Science, University of Colorado Boulder (\$1,000).
- 2024 **Conference Travel Grant**
Awarded by the Graduate and Professional Student Government, University of Colorado Boulder (\$500).
- 2024 **Gary Marsden Travel Award**
Awarded by ACM SIGCHI to attend CHI 2024 (\$3,000).
- 2023 **Best Master's Thesis Award**
Awarded to one of two master's theses in the Department of Industrial Design, KAIST.
- 2022 **Best Paper Award**
Awarded at HCI Korea 2022 [C2].
- 2022 **Top Research Award**
Awarded at the joint research seminar of Seoul National University, KAIST, Sogang University, and Korea University [X5].
- 2021 **Excellence Award**
Awarded at the Korea Metaverse Developer Contest 2021 [X5].
- 2021 **iF Design Award (Winner, UI/UX) and IDEA (Bronze, Digital Interaction)**
Contributed the information architecture to the control UI/UX of the Mobile Clinic Module, a negative-pressure isolation ward (team award, a multi-lab project of the Department of Industrial Design, KAIST).
- 2021 **Honorable Mention Award**
Awarded to the top 5% of submissions at CHI 2021 [C1].
- 2016 **Top Award**
Graduated first in class from the Maya 3D computer graphics course, Green Computer Academy ([portfolio video](#)).
- 2016 **Alumni Award**
Graduated second in class across all departments of the College of Fine Arts, Seoul National University.

Invited Talks and Lectures

Summary: I gave six talks and guest lectures on generative video, content creation, AI tools, prototyping, and moving from art and film VFX to VR at Fujitsu Research of America, KAIST, Seoul National University, the University of Colorado Boulder, and Kookmin University.

- October 2025* **Fujitsu Research of America Lunch Seminar**
Grounding Generative Video in Real-World Geographies for Spatial Consistency
- May 2025* **KAIST Make Lab Seminar**
Reimagining How We Create and Consume Content

- May 2025* **Seoul National University Human-Centered Computing Systems Lab Seminar**
Reimagining How We Create and Consume Content
- February 2025* **University of Colorado Boulder, CSCI 7000 How to Hack Almost Anything Guest Lecture**
Utilizing AI Services and APIs for Coding and Virtual Agent Creation
- October 2022* **KAIST ID220 Interaction Prototyping Class Guest Lecture**
DIY Arduino paper cases without 3D printing, using Adobe Illustrator, Autodesk Fusion 360, and Blender
- November 2019* **Kookmin University Department of Entertainment Design Career Seminar**
Transitioning from Art and Film VFX to VR Content Creation

Teaching and Mentoring

Summary: I have taught two courses as a teaching assistant and mentored four undergraduate researchers.

- Fall 2025* **Teaching Assistant**
CSCI 1300 Starting Computing (C++), Instructor: Rhonda Hoenigman
- Fall 2022* ID220 Interaction Prototyping (Arduino)
- Spring 2024* **Undergraduate Research Assistant**
Chan Hu Wie, Yejin Jang, Dong-Uk Kim, Yurim Son (Chung-Ang University)

Service

Summary: I review for CHI and DIS and mentor prospective and international students.

- 2025 – Present* **Reviewer**
CHI 2025-2026, DIS 2026
- Outreach and Leadership**
- 2025* University of Colorado Boulder Prospect Affiliate Match Program, helping prospective students prepare their PhD applications
- 2012* Seoul National University Mentoring Program, helping international students adjust to campus life

Media Coverage

Summary: My Google Ph.D. Fellowship and my research on bimanual haptics in VR were covered by the University of Colorado Boulder and Microsoft Research.

- October 2025* **University of Colorado Boulder Graduate School News**
[“CU Boulder graduate student named a Google PhD fellow”](#)
- May 2021* **Microsoft Research Blog**
[“Microsoft Research collaborates with KAIST in Korea to explore bimanual interactions with haptic feedback in virtual reality”](#)

Film VFX [\(showreel\)](#)

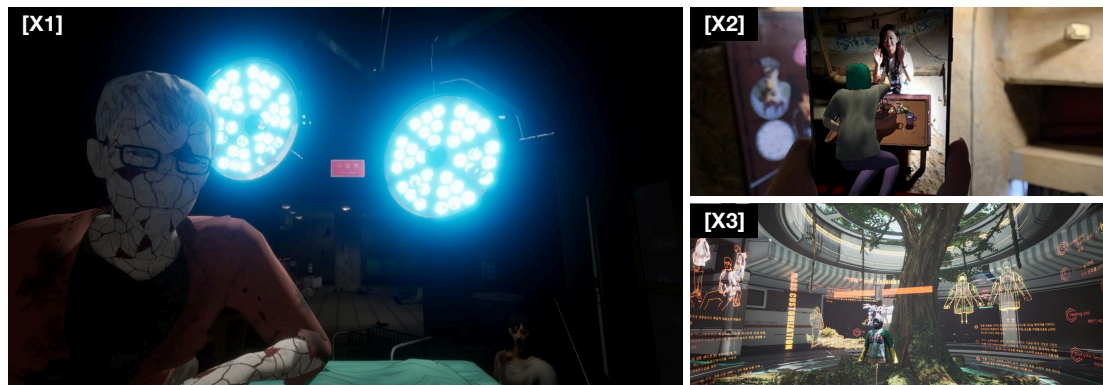
Summary: I worked as a VFX compositor on six feature films [F1-F6]; three of them each drew over five million viewers in South Korea.



- 2018 [F6] <Along with The Gods: The Last 49 Days> ([VFX showreel](#)), Director: Yong-Hwa Kim.
2018 Grand Bell Awards - Best Visual Effects, over 12 million viewers in South Korea alone as of 2024.
- 2017 [F5] <1987: When The Day Comes> ([VFX showreel](#)), Director: Jun-Hwan Jang. *2018 Blue Dragon Film Awards - Top Film Award, over 5 million viewers in South Korea alone as of 2024.*
- 2017 [F4] <Along with The Gods: The Two Worlds> ([VFX showreel](#)), Director: Yong-Hwa Kim.
2018 Blue Dragon Film Awards - 4 Crowns, including Technology Award, over 11 million viewers in South Korea alone as of 2024.
- 2017 [F3] Kung Fu Yoga ([VFX showreel](#)), Director: Stanley Tong
- 2017 [F2] <Real> ([trailer](#)), Director: Sa-Rang Lee
- 2017 [F1] <Fabricated City> ([trailer](#)), Director: Kwang-Hyun Park

Immersive Content [\(showreel\)](#)

Summary: I created VR films, AR exhibitions, and AR apps as a CG generalist [X1-X5], developer [X3], and camera operator [X4].

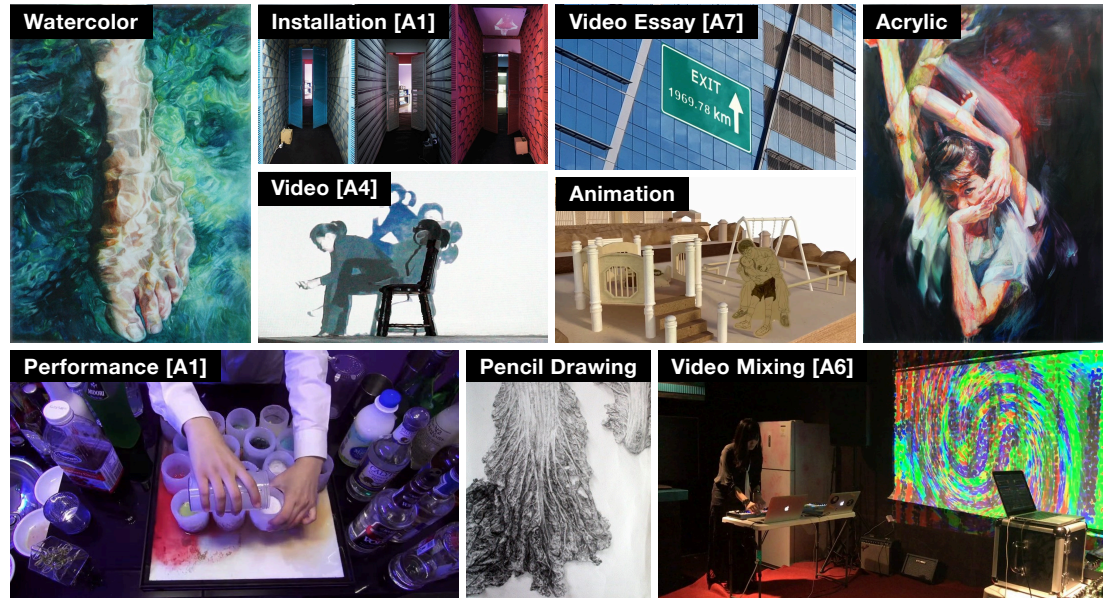


- 2021 [X5] VR Boxing Game <Meta-Boxing> ([video](#)), Supervisor: Woontack Woo

- 2019 [X4] AR Mobile App <LGU+ 5G AR> ([video](#), [article](#)), Supervisor: Sun-Gu Kim
- 2019 [X3] VR Exhibition <Fashion For Help> ([video](#)), Supervisor: Young-Mo Son
- 2019 [X2] AR Exhibition <The Tide>, Supervisor: Sang-Hyoun Lee
- 2019 [X1] VR Toon Film <The Tide> ([Steam](#)), Director: Tae-Kyung Yoo. *Officially invited to the "New Frontier" category at the 2019 Sundance Film Festival.*

Art Exhibitions and Live Performances ([showreel](#))

Summary: I took part in art events as an artist [A1-A5, A7] and video jockey [A6].



- 2018 [A7] Group Exhibition <Lapses>, Platform-L Contemporary Art Center, Curator: Eobchae (funded by Hyundai's ZER01NE project)
- 2016 [A6] Live Video Jockey Performance <We Play>, Sangsangmadang, Supervisor: Hoon-Gyu Park (Parkpunk)
- 2016 [A5] Invited Exhibition <Don't fake it, believe it>, Myhtake Museum
- 2015 [A4] Group Exhibition <Dirt Luv for Graduation> (videos: [Package for Me](#), [Human-furniture](#), [Walking on the Spot](#)), Seoul National University
- 2015 [A3] Group Exhibition <8-bit>, Seoul National University
- 2015 [A2] Group Exhibition <Crawling>, Seoul National University
- 2014 [A1] Group Exhibition <The Great Exodus> (videos: [Closet Inside the Closet](#), [1:1 Bar](#)), Seoul National University. *One of three finalists for the Brighton Prize.*